

		Parul University Faculty of Engineering and Technology Parul Institute of Engineering and Technology Department: AI-ML/AI-RO/AI/AI-DS/CSE/MICRO/SAP/QUICK/ORACLE/IT/AERO		
Subject Name	PROBABILITY, STATISTICS AND NUMERICAL METHODS	A.Y	2025/2026	
Subject Code	303191251	Semester	4th	
Tutorial-1				
Sr No	Question	COs	B.T	Competence
1	The following data gives X: 10, 12, 13, 16, 17 and Y: 20, 22, 23, 26, 27. Calculate Karl Pearson’s correlation coefficient and interpret the result.	1	3	Apply
2	Compute the regression equation of Y on X for the data: X = (5, 7, 9, 11, 13), Y = (15, 19, 23, 25, 29).	1	4	Analyze
3	For the following paired observations (X,Y): (2,5), (3,6), (4,8), (5,10), find the regression lines of X on Y and Y on X.	1	4	Analyze
4	Determine Spearman’s Rank Correlation coefficient for the ranks given by two judges: Judge A → 1, 3, 4, 2, 5; Judge B → 2, 1, 5, 3, 4.	1	3	Apply
5	Fit a straight line of the form y = a + bx for data: x = (0,1,2,3), y = (1,2,2,4).	1	5	Evaluate
6	Fit a second-degree parabola y = a + bx + cx² for the data: x = (1,2,3,4), y = (2,5,10,17).	1	6	Create
7	Given the following data X = (2, 4, 6, 8, 10), Y = (4, 5, 7, 10, 14), calculate covariance and conclude the type of relationship.	1	2	Understand
8	The correlation between X and Y is 0.85. If $\sigma_x = 3$ and $\sigma_y = 4$, compute the regression coefficient b_{yx} .	1	3	Apply
9	For the data X: 12, 15, 17, 20, 22 and Y: 30, 35, 40, 45, 50; determine the coefficient of determination (R ²) and interpret.	1	5	Evaluate
10	For the dataset X: 1, 2, 3, 4, 5 and Y: 2, 5, 3, 8, 7, check whether linear curve fitting is suitable by plotting trend (conceptual).	1	2	Understand